Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS at USC with 2+ years building operational software, SQL-backed reporting, data integrations, and tested workflows. Strong in system design, GUI-backed tools, troubleshooting, documentation, and reliable software delivery for infrastructure-oriented environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and optimized SQL-backed operational workflows supporting 500+ users, reducing latency from **1.2s to 350ms** through requirements analysis, query tuning, and service fixes.
- Built reporting and validation pipelines processing 5,000+ monthly jobs with reconciliation checks, CI/CD releases, and runbooks, improving data-transfer reliability and supportability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React and Flask applications handling 10,000+ daily requests, translating product requirements into GUI workflows, REST APIs, tests, and maintainable software components.
- Prioritized defects, edge cases, and user feedback across document-processing workflows, reducing turnaround by **60%** while improving release confidence and technical support clarity.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python evaluation pipelines across 100,000+ records, producing reproducible reports, debugging artifacts, and visual summaries that clarified model behavior and data quality.
- Analyzed failure patterns across generated outputs and datasets, translating complex findings into actionable recommendations while strengthening validation coverage and technical decision-making.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting scalable integrations and reliable production workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and accelerating defect isolation across services.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based platform with WebSockets and REST APIs, supporting 2M+ synchronized records and six workflows through modular services and responsive interfaces.
- Coordinated Agile delivery with QA and business teams, automating Docker-based build and test flows that cut release time by **50%** and improved deployment consistency.

Skills

Software: Python, Java, TypeScript, JavaScript, SQL, C++, React, Flask, FastAPI, Node.js, REST APIs
Systems & Operations: System design, GUI applications, reporting tools, data transfer, troubleshooting, maintenance procedures, training materials
Utility Context: SCADA/data-interface familiarity, CAISO and energy-market learning, compliance awareness, change management, operational logic
Data & Delivery: PostgreSQL, MySQL, SQL Server, MongoDB, ETL, Git/GitHub, Docker, AWS, Jenkins, CI/CD, unit and integration testing

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, combining schema inspection, query generation, Dockerized services, and debugging flows for faster data validation.

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented scheduling algorithms and visual comparisons for operating-system workloads, reinforcing operational logic, C++ fundamentals, data structures, and clear explanation of system behavior.

HMIS Code | *MERN, TypeScript, ABDM APIs, Sentry*
- Built a management platform integrating user, provider, and claim workflows with secure APIs, structured data validation, monitoring, and reporting-oriented operational reliability.